

analysis

the model $y = \beta_0 + \beta_1 \text{edu} + \varepsilon$
dependent and 'Educational Level' as

Plotting the regression line

- Drag 'Current Salary' into the vertical axis box and 'Beginning Salary' in the horizontal axis box.
- Click 'Fit' bar. Make sure the Method is regression in the Fit box. Then click 'OK'.

population

N

n

μ

μ

σ

MCA

$$\sigma_1 = \frac{\sigma}{\sqrt{n}}$$

$N > 0.05$, it may be best to use the

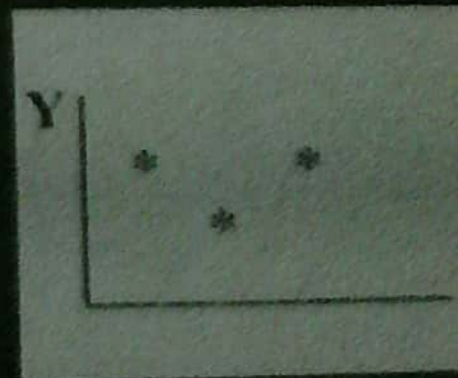
Correlation

Finding the relationship between two quantitative variables without being able to infer causal relationships

Correlation is a statistical technique used to determine the degree to which two variables are related

Scatter diagram

- Rectangular coordinate
- Two quantitative variables
- One variable is called independent (X) and the second is called dependent (Y)
- Points are not joined
- No frequency table



Important Terms

Sporadic: disease occurs occasionally, irregularly ■

Endemic: disease stays in population at low frequency ■

Epidemic: sudden outbreak in disease above typical level ■

Pandemic: epidemic over wide area (may be entire world). ■

Morbidity: all reported cases of disease, illness, and disability ■

Mortality: reported deaths due to a disease ■

ROC CURVES

Receiver Operating Characteristic (ROC) Curve, is a plot of the true positive rate against the false positive rate for the different possible cut points of the diagnostic test.

VES

0.90-1 = excellent (A)

0.80-0.90 = good (B)

0.70-0.80 = fair (C)

0.60-0.70 = poor (D)

0.50-0.60 = fail (F)

M.C.B.

ROC CURVES

↑ *ROC*
ROC analysis is a part of field called "Signal Detection Theory"

Which was developed during World War II for the analysis of radar images. *ROC*

It was not until 1970's that signal detection theory was recognized as useful for interpreting medical test results. *ROC*

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Binomial Distribution - II

Mean, Variance and Standard Deviation of the Binomial Distribution

$$\checkmark \text{ Mean} = E(X) = \sum_{x=0}^n xP(X) = np \checkmark$$

$$\checkmark \text{ Variance} = E(X^2) - [E(X)]^2 = npq \checkmark$$

$$\checkmark \text{ SD} = \sqrt{\text{variance}} = \sqrt{npq} \checkmark$$

$$n = 5$$

$$p = 0.60$$

$$1 - p = q = 0.40$$

$$\text{Mean} = 5 \times 0.60 = 3$$

$$\text{Variance} = 5 \times 0.60 \times 0.40 = 1.2$$

$$\text{SD} = 1.90$$

Binomial Distribution - II

Interpretation of the results

It is especially important to interpret results. The Range Rule of Thumb suggests that values are unusual if they are outside of these limits.

Maximum Usual Value = $\mu + 2\sigma$

Minimum Usual Value = $\mu - 2\sigma$

The Discrete Distribution used for such instances is called Poisson Probability Distribution.

Named after a French mathematician Simeon Denis Poisson



https://en.wikipedia.org/wiki/Simeon_Denis_Poisson

1960

Norman H. Nie

Software Program

& Collaboration

Introduction: What is SPSS?

Lecture no 76

- Originally it is an acronym of Statistical Package for the Social Science but now it stands for Statistical Product and Service Solutions

Statistical Package of Social Sciences

Question: What is SPSS?

Lecture no 76

SPSS is an acronym of Statistical Package of the Social Science but now it is Statistical Product and Service

It is the most popular statistical software which can perform highly

The Four Windows:

Data editor

Output viewer

Syntax editor

Script window

The Four Windows: Data Editor

- Data Editor

Spreadsheet-like system for defining, entering, editing, and displaying data. Extension of the saved file will be "sav".

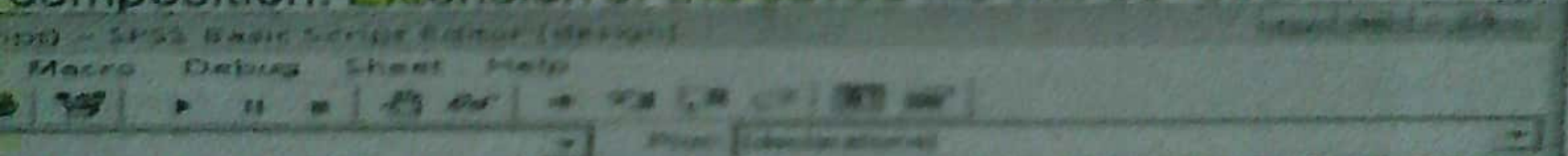
SPSS Data Editor

The Four Windows: Output Viewer

- Output Viewer

Displays output and errors. Extension of the saved file will be ".spv."

in a BASIC-like language. Text editor for syntax composition. **Extension of the saved file will be "sbs."**

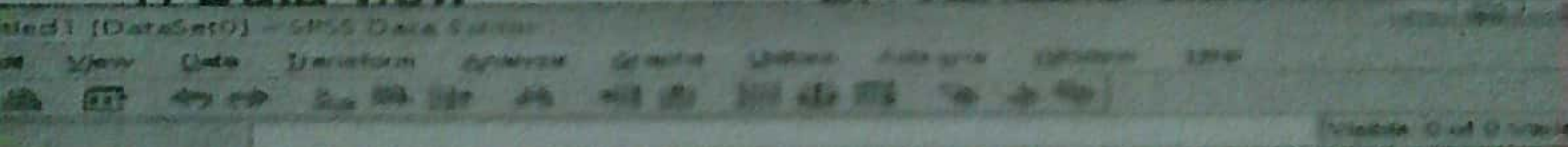


Opening SPSS

- The default window will have the data editor
- There are two sheets in the window:

1. Data view

2. Variable view



Variable View window: Decimals

- Decimals
 - Number of decimals
 - It has to be less than or equal to 16

Variable View window: Label

- Label
 - You can specify the details of the variable
 - You can write characters with spaces up to 256 characters

Defining the value labels

- Click the cell in the values column as shown below
- For the value, and the label, you can put up to 50 characters.
- After defining the values click add and then click OK

